BM20A6100 Advanced Data Analysis and Machine Learning

Exercise 2: Visualizing Neural Networks

Deadline: 08.11.2026

Submitting the solutions

Create a PDF containing your answers. Include all the relevant equations. Feel free to use graphs/figures.

AI usage

*AI tools may be used as a programming assistant. It however, should **NOT** perform data analysis, create visualizations, interpret the results, or write any part of the report*

Task (8 points)

One way to inspect neural networks is to visualize the hidden representations. Since the layers have usually a high number of dimensions (the # of nodes in the layer), dimensionality reduction techniques need to be used for the visualization.

Train a neural network classifier for the MNIST digit dataset (`torchvision.datasets.MNIST()`). You can use any architecture you like.

Use **PCA, LLE, UMAP, and SOM** to visualize **the input data**, **at least one hidden representation**, and **the latent representation before the classifier head**.

Analyze the results. Can you see the classes separated? Do some DR methods work for this visualization better than some others and why? What happens if you don't train the model?

Return a 1-4 page report to the Moodle submission tool.